

EX.NO : 01

FAMILIARIZING WITH WINDOWS NETWORK COMMANDS

AIM :

To familiarize with windows network commands and their outputs.

PROCEDURE :

1. Open the Command prompt by typing “CMD” in the Run Dialogue.
2. Once the Command prompt opens type the commands.

COMMAND DESCRIPTION :

Sl.NO	COMMAND	USE
1)	Ipconfig	This command can be utilized to verify a network connection as well as verify your network settings.
2)	Netstat	Displays active a TCP connections, ports on which the computer is listening, Ethernet statistics, the IP routing table etc...
3)	Tracert	The tracert command is used to visually see a network packet being sent and received and the amount of hops required for that packet to get to its destination.
4)	Ping	Helps in determining TCP/IP networks ip address as well as determine issues with the network and assists in resolving them.
5)	Pathping	Provides information about network latency and network loss at intermediate hops between a source and destination pathping sends.

6)	Nslookup	Displays information that you can use to diagnose Domain Name System (DNS) infrastructure.
7)	Nbtstat	MS_DOS utility that displays protocol statistics & current TCP/IP connections using NBT.
8)	Getmac	DOS command used to show both local & remote MAC addresses when run with no parameters (i.e. getmac) it displays MAC addresses for the local system. When run with the /s parameter (Eg. Getmac /s \\too> it displays MAC address for the remote computer).

OUTPUT :

Ipconfig

```

C:\Windows\system32\cmd.e  X  +  v
C:\Users\MSD>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    IPv6 Address. . . . . : 2401:4900:2325:1cd1:bf17:7e12:afed:6a20
    Temporary IPv6 Address. . . . . : 2401:4900:2325:1cd1:f47a:5620:6663:c43f
    Link-local IPv6 Address . . . . . : fe80::3d7d:8789:4d20:4a40%16
    IPv4 Address. . . . . : 192.168.153.172
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::c022:3dff:fe0c:fe56%16
                                192.168.153.211

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

```

Netstat

```
C:\Windows\system32\cmd.e: X + v
Microsoft Windows [Version 10.0.22621.3155]
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C:\Users\MSD>netstat

Active Connections

Proto Local Address Foreign Address State
TCP 192.168.153.172:50729 20.198.119.84:https ESTABLISHED
TCP 192.168.153.172:50807 152.195.38.76:http TIME_WAIT
TCP 192.168.153.172:50809 152.195.38.76:http TIME_WAIT
TCP [2401:4900:2325:1cd1:f47a:5620:6663:c43f]:50797 g2600-140f-f400-0000-0000-0000-1730-e221:http TIME_WAIT
TCP [2401:4900:2325:1cd1:f47a:5620:6663:c43f]:50798 g2600-140f-f400-0000-0000-0000-1730-e221:http TIME_WAIT
TCP [2401:4900:2325:1cd1:f47a:5620:6663:c43f]:50800 g2600-140f-f400-0000-0000-0000-1730-e221:http TIME_WAIT
```

Tracert

```
C:\Windows\system32\cmd.e: X + v
C:\Users\MSD>tracert www.srmist.edu.in

Tracing route to srmist-alb-630144276.ap-south-1.elb.amazonaws.com [35.154.166.26]
over a maximum of 30 hops:

 1  3 ms  3 ms  3 ms  192.168.153.211
 2  *      *      *      Request timed out.
 3  *      *      *      Request timed out.
 4  46 ms  30 ms  22 ms  10.50.221.46
 5  59 ms  40 ms  22 ms  125.19.176.89
 6  74 ms  29 ms  66 ms  182.79.239.197
 7  *      *      *      Request timed out.
 8  *      *      *      Request timed out.
 9  162 ms  82 ms  81 ms  52.95.64.170
10  86 ms  94 ms  76 ms  52.95.64.167
11  103 ms  70 ms  73 ms  52.95.66.127
12  84 ms  74 ms  79 ms  52.95.65.133
13  *      *      *      Request timed out.
14  *      *      *      Request timed out.
15  *      *      *      Request timed out.
16  *      *      *      Request timed out.
17  *      *      *      Request timed out.
18  *      *      *      Request timed out.
19  *      *      *      Request timed out.
20  *      *      *      Request timed out.
21  *      *      *      Request timed out.
22  *      *      *      Request timed out.
23  *      *      *      Request timed out.
24  *      *      *      Request timed out.
25  *      *      *      Request timed out.
26  *      *      *      Request timed out.
27  *      *      *      Request timed out.
28  *      *      *      Request timed out.
29  *      *      *      Request timed out.
30  *      *      *      Request timed out.

Trace complete.
```

Ping

```
C:\Windows\system32\cmd.e: X + v
Microsoft Windows [Version 10.0.22621.3155]
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C:\Users\MSD>ping 1.1.1.7

Pinging 1.1.1.7 with 32 bytes of data:
Reply from 1.1.1.7: bytes=32 time=279ms TTL=56
Reply from 1.1.1.7: bytes=32 time=44ms TTL=56
Reply from 1.1.1.7: bytes=32 time=84ms TTL=56
Reply from 1.1.1.7: bytes=32 time=73ms TTL=56

Ping statistics for 1.1.1.7:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 44ms, Maximum = 279ms, Average = 120ms
```

Pathping

```
C:\Users\MSD>pathping www.srmist.edu.in

Tracing route to srmist-alb-630144276.ap-south-1.elb.amazonaws.com [13.127.51.112]
over a maximum of 30 hops:
  0  MSD-K67DBKG [192.168.153.172]
  1  192.168.153.211
  2  * * *
Computing statistics for 25 seconds...
Hop  RTT      Source to Here   This Node/Link   Address
  0                               Lost/Sent = Pct  Lost/Sent = Pct  MSD-K67DBKG [192.168.153.172]
  1    7ms      0/ 100 = 0%      0/ 100 = 0%      |
  1    7ms      0/ 100 = 0%      0/ 100 = 0%      192.168.153.211

Trace complete.
```

Nslookup

```
C:\Users\MSD>nslookup www.srmist.edu.in
Server:  UnKnown
Address:  192.168.153.211

Non-authoritative answer:
Name:      srmist-alb-630144276.ap-south-1.elb.amazonaws.com
Addresses:  13.127.51.112
            35.154.166.26
Aliases:    www.srmist.edu.in
```

Nbtstat

```
C:\Windows\system32\cmd.e: X + v
Microsoft Windows [Version 10.0.22621.3155]
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C:\Users\MSD>nbtstat

Displays protocol statistics and current TCP/IP connections using NBT
(NetBIOS over TCP/IP).

NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
          [-r] [-R] [-RR] [-s] [-S] [interval] ]

-a (adapter status) Lists the remote machine's name table given its name
-A (Adapter status) Lists the remote machine's name table given its
                        IP address.
-c (cache)           Lists NBT's cache of remote [machine] names and their IP addresses
-n (names)           Lists local NetBIOS names.
-r (resolved)        Lists names resolved by broadcast and via WINS
-R (Reload)          Purges and reloads the remote cache name table
-S (Sessions)        Lists sessions table with the destination IP addresses
-s (sessions)        Lists sessions table converting destination IP
                        addresses to computer NETBIOS names.
-RR (ReleaseRefresh) Sends Name Release packets to WINS and then, starts Refresh

RemoteName  Remote host machine name.
IP address   Dotted decimal representation of the IP address.
interval     Redisplays selected statistics, pausing interval seconds
              between each display. Press Ctrl+C to stop redisplaying
              statistics.

C:\Users\MSD>nbtstat -n

Ethernet:
Node IpAddress: [0.0.0.0] Scope Id: []

    No names in cache

Bluetooth Network Connection:
Node IpAddress: [0.0.0.0] Scope Id: []

    No names in cache

Wi-Fi:
Node IpAddress: [192.168.153.172] Scope Id: []

    NetBIOS Local Name Table

    Name                Type                Status
    -----
    MSD-K67DBKG          <20>    UNIQUE          Registered
    MSD-K67DBKG          <00>    UNIQUE          Registered
    WORKGROUP             <00>    GROUP           Registered

Local Area Connection* 1:
Node IpAddress: [0.0.0.0] Scope Id: []

    No names in cache

Local Area Connection* 2:
Node IpAddress: [0.0.0.0] Scope Id: []

    No names in cache
```

Getmac

```
C:\Windows\system32\cmd.e: X + v
Microsoft Windows [Version 10.0.22621.3155]
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C:\Users\MSD>getmac

Physical Address      Transport Name
=====
3C-91-80-54-6C-9F     \Device\Tcpip_{97E7618B-4E96-47F3-BFE3-AB234CAED9EB}
9C-5A-44-40-C4-ED     Disconnected
3C-91-80-54-6C-A0     Media disconnected
```